

Buyun Liang

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EDUCATION

University of Pennsylvania

Ph.D. in Computer and Information Science | Advisor: Prof. René Vidal

Philadelphia, PA, USA

Sep 2023 - Present

University of Minnesota, Twin Cities

M.Sc in Computer Science | Advisor: Prof. Ju Sun

Minneapolis, MN, USA

Sep 2020 - May 2023

University of Minnesota, Twin Cities

M.Sc in Materials Science (Ph.D. Track) | Advisor: Prof. Ilja Siepmann

Minneapolis, MN, USA

Sep 2018 - Aug 2020

Nanjing University

B.Sc in Physics (Elite Program)

Nanjing, Jiangsu, China

Sep 2014 - Jun 2018

PUBLICATIONS

Trustworthy AI: Safe & Reliable LLMs.....

- [1] **REALISTA: Realistic Latent Adversarial Attacks that Elicit LLM Hallucinations**
Buyun Liang, Jinqi Luo, Liangzu Peng, Kwan Ho Ryan Chan, Darshan Thaker, Kaleab A Kinfu, Fengrui Tian, Hamed Hassani, René Vidal
ICML 2026 [[paper](#)][[code](#)]
- [2] **SECA: Semantically Equivalent and Coherent Attacks for Eliciting LLM Hallucinations**
Buyun Liang, Liangzu Peng, Jinqi Luo, Darshan Thaker, Kwan Ho Ryan Chan, René Vidal
NeurIPS 2025 [[paper](#)][[code](#)]
- [3] **KDA: A Knowledge-Distilled Attacker for Generating Diverse Prompts to Jailbreak LLMs**
Buyun Liang, Kwan Ho Ryan Chan, Darshan Thaker, Jinqi Luo, René Vidal
arXiv preprint, 2025 [[paper](#)]

Optimization for AI.....

- [4] **NCVX: A General-Purpose Optimization Solver for Constrained Machine and Deep Learning.**
Buyun Liang, Tim Mitchell, Ju Sun.
NeurIPS 2022 Workshop: Optimization for Machine Learning [[paper](#)][[website](#)][[code](#)][[talk](#)]
- [5] **NCVX: A User-Friendly and Scalable Package for Nonconvex Optimization in Machine Learning.**
Buyun Liang, Tim Mitchell, Ju Sun.
arXiv preprint, 2021 [[paper](#)]

Trustworthy AI: Adversarial Robustness in Vision.....

- [6] **Optimization and Optimizers for Adversarial Robustness**
Hengyue Liang, Buyun Liang, Le Peng, Ying Cui, Tim Mitchell, Ju Sun
arXiv preprint, 2023 [[paper](#)]
- [7] **Optimization for Robustness Evaluation beyond ℓ_p Metrics**
Hengyue Liang, Buyun Liang, Ying Cui, Tim Mitchell, Ju Sun
ICASSP 2023 & NeurIPS 2022 Workshop: Optimization for Machine Learning [[paper](#)][[poster](#)]
- [8] **Implications of Solution Patterns on Adversarial Robustness**
Hengyue Liang, Buyun Liang, Ju Sun, Ying Cui, Tim Mitchell
CVPR 2023 Workshop: Art of Robustness [[paper](#)]

AI for Science & Scientific Computing.....

- [9] **Neural Topology Optimization Based on Differential Programming with Principled Constrained Optimization**
Ryan de Vera, Buyun Liang, Binyao Guo, Qizhi He, Ju Sun
ASME 2024 International Mechanical Engineering Congress and Exposition [[paper](#)]
- [10] **Large-scale molecular dynamics simulations of bubble collapse in water: Effects of system size, water model, and nitrogen**
Jingyi L Chen, Jesse L Prelesnik, Buyun Liang, Yangzesheng Sun, Mrugank Bhatt, Christopher Knight, Krishnan Mahesh, J Ilja Siepmann
The Journal of Chemical Physics (JCP Editors' Choice 2023) [[paper](#)]
- [11] **Effect of Non-Condensable Gas on the Thermophysical Properties of Bubbly Water and on Bubble Collapse Dynamics Probed by Molecular Simulations.**
J. Ilja Siepmann, Jingyi L Chen, Buyun Liang, Krishnan Mahesh
The 33rd Symposium on Naval Hydrodynamics, 2020 [[paper](#)]
- [12] **FAIR Framework for Physics-Inspired AI in High Energy Physics (Final Technical Report)**
Roger W Rusack, Priscilla Cushman, Andrew Furmanski, Mathew Fritts, Bhargav Joshi, Taihui Li, Buyun Liang, Joseph Muse, Ju Sun
University of Minnesota Technical Report, 2024 [[paper](#)]

TALKS

REALISTA: Realistic Latent Adversarial Attacks that Elicit LLM Hallucinations <i>International Conference on Machine Learning (ICML)</i>	2026
SECA: Semantically Equivalent and Coherent Attacks for Eliciting LLM Hallucination <i>Neural Information Processing Systems (NeurIPS)</i>	2025
Deep Learning with Nontrivial Constraints <i>SIAM International Conference on Data Mining</i>	2023
When Deep Learning Meets Nontrivial Constraints <i>Midwest Machine Learning Symposium</i>	2023
Toward Trustworthy AI — Robustness and Beyond <i>Midwest Machine Learning Symposium</i>	2023
When Deep Learning Meets Nontrivial Constraints <i>3M poster session</i>	2023
NCVX: A General-Purpose Optimization Solver for Constrained Machine and Deep Learning <i>NeurIPS Workshop: Optimization for Machine Learning</i>	2022

PROFESSIONAL SERVICE

Organizer.....

Supported the organization and coordination of oral, tutorial, and keynote poster sessions
@ 2025 Conference on Lifelong Learning Agents

Led the organization of a 2-Hour Tutorial "Deep Learning with Nontrivial Constraints",
@ 2023 SIAM International Conference on Data Mining

Reviewer.....

International Conference on Machine Learning (ICML)	2026
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Computer Vision and Pattern Recognition (CVPR)	2026
Neural Information Processing Systems (NeurIPS)	2025-2026
International Conference on Learning Representations (ICLR)	2025-2026
International Conference on Acoustics, Speech, and Signal Processing (ICASSP)	2024-2026
Uncertainty in Artificial Intelligence (UAI)	2023-2026
International Joint Conference on Neural Networks (IJCNN)	2025-2026
IEEE International Workshop on Machine Learning for Signal Processing (MLSP)	2025-2026
Artificial Intelligence and Statistics (AISTAT)	2023
NeurIPS 2023 Workshop: Optimization for Machine Learning (OPT)	2023

(Update: June 5, 2026)